Poisoning in Children

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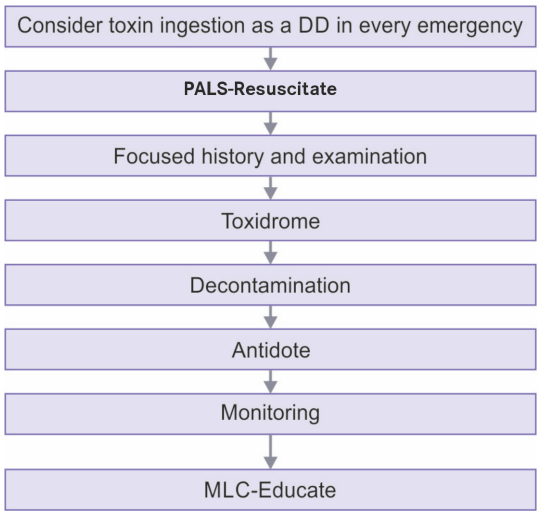
Introduction

Poisonings continue to baffle pediatricians in India because of the sheer variety of poisonous substances and the myriad clinical manifestations of exposure to these agents.

What this article proposes to do is to:

- ☑ Recognize the symptoms wherein poisoning can be considered as a differential diagnosis (DD)
- ☑ Identify common household poisonings that most of us are likely to encounter
- ☑ Identify the most likely poison based on the symptoms and signs
- ☑ Initial stabilization of such children
- ☑ General principles of management
- ☑ Specific management.

Key Components in the Management of Poisoning



A pediatrician in office practice often needs to quickly decide based on history and signs seen whether the child in front is showing signs of poisonings.

Age group	Mental status	History of substance consumed
Toddler	Explorative	Available—household
Adolescent	Experiential/disturbed	Not available—recreational/suicidal

Diagnosis

In most of the situations, a child with poisoning will present like an unknown emergency.

When to suspect poisoning?

- ☑ Sudden onset of altered consciousness or breathing difficulty or convulsion.
- ☑ There is a vacuum in the history, e.g., apparently well child playing in the room unsupervised, found drowsy, or convulsing after few hours. No one knows about this few hours.
- ☑ In any emergency, i.e., intoxication after injury, infection
- ☑ If toxin ingestion is strongly suspected, it is mostly accidental in young kids and intentional with suicide in an adolescent child.

What should the pediatrician ask, when the child is brought with definite history of poisoning?

Questions		***Intervention***	***Contraindication***	***Example***
When was the poison consumed?	<1 hour since ingestion	Elimination stomach wash can be offered	When it is a hydrocarbon/ volatile substance like kerosene	Antidepressants, and lithium
How much quantity of the substance was consumed?	Helps decide toxic dose			Toxic dose varies for various drugs
Did the parents use any techniques to make the child forcefully vomit out the poison?	Home remedy		Can cause aspiration pneumonias especially when used with hydrocarbons and volatile substances	
What were the symptoms which brought it to attention of parents and caregivers?	Helps establish time since drug ingestion based on drug pharmaco-kinetics			Iron poisoning presents in four phases

Resuscitation

Steps of managing a poisoned child follow the ABCDE approach:

Parameter	*Intervention*	*example*
Airway	☑ Suctioning—bronchorrhea and salivation ☑ Edema with corrosives ☑ Early intubation	☑ OP poisoning ☑ Burns inhalational
Breathing	☑ Kussmaul breathing with metabolic acidosis ☑ Bradypnea due to central nervous system (CNS) depressants ☑ Mechanical ventilation	☑ Salicylate poisoning ☑ Tricyclic antidepressant and opiates
Circulation	☑ Bradycardia ☑ Hypotension ☑ Fluid resuscitation	Beta-blocker
Decontamination	Remove clothes contaminated with pesticides	Organophosphate (OP) poisons can be absorbed via skin, wear gloves when doing so
Disability	Alert, verbal, pain, and unresponsive (AVPU) scale for rapid sensorium assessment and pupils to be checked	Various toxidromes associated with pupillary abnormalities
Exposure	Other trauma due to fall after consumption of poison	☑ Common with alcohol ☑ Venipuncture sites for illicit drugs

Toxidrome

Toxidromes for Rapid Recognition

Based on the knowledge of the toxins known for pharmacological or toxicity profile one can make a reasonable guess about the most likely cause of poisoning. A set of symptoms and signs that are classical to shortlist the most likely agent is called a toxidrome.

How to identify the toxidrome based on heart rate (HR), respiratory rate (RR), pupillary changes, and peripheral oxygen saturation-arterial oxygen saturation (SpO_2-SaO_2)?

☑ *Check HR*:

- Heart rate high → patient hot and dry → anticholinergic toxidrome, Datura and TCA (Tricyclic Antidepressants)
- Heart rate high → patient sweating → sympathomimetic
- Heart rate low → cholinergic → organophosphorus
- Blood pressure (BP) and RR low → opiate toxidrome

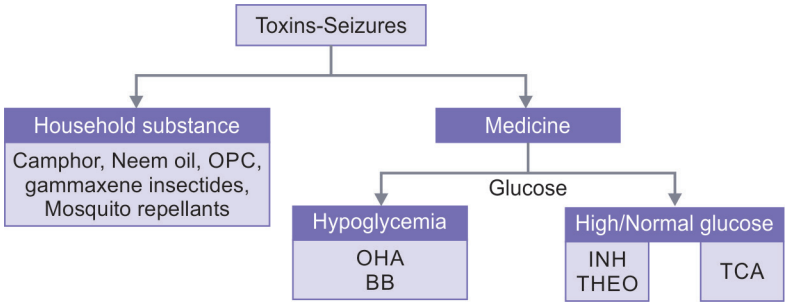
☑ *Check respiratory rate*:

- *Slow and shallow*: CNS depressants and opiates
- *Rapid and deep (effortless tachypnea)*: Metabolic acidosis: iron and salicylate
- *Rapid with retractions*: Hydrocarbon pneumonia and organophosphate compound (OPC)

☑ *Pupil*:

- *Constricted*: OPC, opiate, and phenobarbitone
- *Dilated*: Datura, TCA, and cocaine

☑ *Seizures*:



☑ *SpO_2: Oxygen saturation*

- Both SpO_2 and SaO_2 are low—aspiration pneumonia due to hydrocarbons, cardiac toxins with pulmonary edema, and CNS depressants
- SpO_2 low but SaO_2 normal—methemoglobinemia due to naphthalene, local anesthetic creams, and dapsone
- Both SpO_2 and SaO_2 are normal or high, but child is very sick—cyanide or carbon monoxide poisoning.

- ☑ *Glucose*: By glucometer
- ☑ *Hypoglycemia*: Oral hypoglycemic agents

Toxidrome	*Mnemonic*	*Clinical findings*	*Due to overdose of*
Anticholinergic	☑ Mad as a hatter ☑ Blind as a bat ☑ Hot as a desert ☑ Dry as a bone ☑ Red as a beet	☑ Confused ☑ Tachycardia ☑ Dilated pupils ☑ No sweating ☑ Dry mouth, not passed urine ☑ Flushed cheeks ☑ Tremors	☑ Atropine ☑ Chlorpheniramine ☑ Loratadine ☑ Domperidone ☑ Imipramine ☑ Fluoxetine ☑ Olanzapine ☑ Datura
Sympathomimetic		☑ Tachycardia ☑ Dilated pupils ☑ Agitation and alert ☑ Hyperthermic ☑ Sweating hypertension ☑ Cardiac arrhythmia ☑ Seizures	☑ Epinephrine ☑ Norepinephrine ☑ Dobutamine ☑ Cocaine amphetamines
Cholinergic		☑ Bradycardia ☑ Sweating ☑ Pinpoint pupils ☑ Excessive salivation ☑ Lachrymation ☑ Loose motions ☑ Urination	☑ OP poisons ☑ Carbamate ☑ Drugs used for myasthenia gravis ☑ Neostigmine ☑ Pilocarpine ☑ Edrophonium
Opioid		☑ Bradypnea ☑ Hypotension ☑ Somnolence ☑ Hypothermia ☑ Injection marks	☑ Morphine ☑ Fentanyl ☑ Vicodin ☑ Opium ☑ Heroin

Decontamination

Activated charcoal is more useful after gastric lavage, except in iron, hydrocarbon, and alcohol. Even in an unconscious child it can be given through nasogastric tube (NGT) 1–2 g/kg as a slurry. Charcoal tablets are not equivalent to this.

Antidotes

Antidotes are available only for few toxins. Every hospital emergency department (ED) should stock antidotes. Antidotes are similar to fire extinguishers. They must be always available. If no occasion arises needing their use, one should be happy.

Antidotes	
☑ Iron	☑ Desferrioxamine
☑ Anticholinergics	☑ Atropine, P2 AM
☑ Narcotics	☑ Naloxone
☑ Warfarin	☑ Vitamin K
☑ Oral antidiabetic	☑ Octreotide
☑ Benzodiazepines	☑ Flumazenii
☑ Paracetamol	☑ N—acetyl cysteine
☑ Methemoglobinemia	☑ Methylene blue
☑ INH	☑ Pyridoxin

Individual Household Poisons and their Management

☑ ***Acids and alkalis***: Corrosives

- *Source*: Dish washers and toilet cleansers
- *Toxicity*: Corrosive injury to the airway, eyes, and GI mucosa
- *Symptoms and signs*: Lip edema, burning sensation, oral ulcers, red eyes, esophagitis, laryngeal edema, and viscus perforation
- *Management*: No lavage and charcoal
- Local decontamination with water or saline symptoms-based treatment
- *Specific antidote*: None

☑ ***Hydrocarbons***:

- *Source*: Kerosene, petrol, spirits, and lamp oil
- *Toxicity*: Highly volatile and easy to aspirate into respiratory system

- *Symptoms and signs*: Local effects similar to caustic agents, respiratory symptoms secondary to aspiration pneumonitis
- *Management*: No lavage, no charcoal, respiratory support
- *Specific antidote*: None

☑ ***Alcohols, perfumes and mouthwashes***:

- *Source*: Methanol, isopropyl alcohol (hand rubs), and ethanol
- *Toxicity*:
  - Local irritation based on route of entry
  - Metabolized by liver to acids and cause severe metabolic acidosis
- *Symptoms and signs*: Nausea, vomiting, CNS depression, hypoglycemia, and visual toxicity
- *Management*: No lavage, charcoal, and hemodialysis if early
- *Specific antidote*: Fomepizole

☑ ***Naphthalene***:

- *Source*: Mothballs, deodorants, and dyes
- *Toxicity*: Inhaled naphthalene vapors cause respiratory toxicity whereas ingested naphthalene causes hemotoxicity
- *Symptoms and signs*: Respiratory symptoms of local irritation and systemic features such as CNS irritation, methemoglobinemia, and renal injury
- *Management*: Close monitoring for a week after exposure as features of toxicity may be delayed
- *Specific antidote*: None, methemoglobinemia is managed as per protocol

☑ ***Bleach***: Sodium hypochlorite

- *Source*: Household bleaching agents and disinfectants
- *Toxicity*: Ingested or inhaled hypochlorite releases hypochlorite acid and chlorine gas
- *Symptoms and signs*: Nausea, vomiting, eye irritation, cough, and esophageal injury
- Management is as for corrosive poisoning
- *Specific antidote*: None

☑ ***Eucalyptus oil***:

- *Source*: Insect repellants, antiseptics, and fragrances
- *Toxicity*: Cineole is the most important component and causes intense local irritation and CNS depression at toxic levels

- *Symptoms and signs*: Irritant effects on GI, respiratory, and skin as per route of entry. Can cause CNS depression.
- *Management*: Only symptomatic treatment is needed
- *Specific antidote*: None

☑ ***Neem oil***:

- *Source*: Disinfectants, mosquito repellent, and antiseptic preparations
- *Toxicity*: Chief toxic component is an azadirachtin, a mitochondrial poison
- *Symptoms and signs*: GI irritation followed by hepatic injury followed by CNS depression, very similar to phosphorus poisoning
- *Management:* Symptomatic treatment only
- *Specific antidote*: None

☑ ***Camphorated oils***:

- *Source*: Rubs, perfumes, pain relief gels and oils, and inhalation capsules
- *Toxicity*:
  - Usually as a mixture with menthol, methyl salicylate, eucalyptus oil in various combinations
  - Irritant and CNS stimulation
- *Symptoms and signs*:
  - Intense local irritation based on route of entry of skin, eyes, GI, and respiratory tract
  - CNS stimulation with seizures
- *Management*: Symptomatic treatment and benzodiazepines
- *Specific antidote*: None

☑ ***Rodenticide (phosphorus)***:

- *Source*:
  - Inorganic phosphorus as yellow phosphorus and ratol
  - Organic phosphorus as zinc or aluminum phosphide
- *Toxicity*: Mainly by ingestion, also locally from fireworks and fertilizers, rarely inhalation. Phosphorus is a protoplasmic poison and can affect GI tract, liver, renal, and CNS
- *Symptoms and signs*: Very similar to paracetamol poisoning in the general scheme of presentation:
  - *Stage 1*: Local gastrointestinal irritation effects, shock, and garlic odor with luminous stools

- *Stage 2*: Apparent wellness lasting 2–3 days
- *Stage 3*: Systemic features characterized by acute liver cell failure, renal impairment, myocardial depression, and CNS symptoms

- *Management*: Careful lavage with dilute potassium permanganate solution immediately after ingestion within 1 hour, followed by close monitoring and supportive therapy especially the hepatobiliary system
- *Specific antidote*: None

☑ ***Pyrethroids***:

- *Source*: Mosquito repellent solutions and coils, *Laxman Rekha*, and cockroach repellents
- *Toxicity*: Mainly by ingestion and local spillage on skin or eyes
- *Pyrethroids are of two types*:
  1. Type 1 allethrin causes sodium channel activation and neuronal excitation
  2. Type 2 deltamethrin, fenvalerate causes hyperpolarization and neuromuscular depression
- *Symptoms and signs*:
  - Mostly local irritation of eyes, skin, and GI mucosa
  - At toxic doses effects of CNS irritation such as tremors, ataxia, and seizures or neuromuscular weakness
- *Management*:
  - Gastric lavage if within 6 hours of ingestion
  - Local soothing agents like calamine, eye wash with sterile saline, and benzodiazepines for CNS excitation
- *Specific antidote*: None

☑ ***Usually harmless***:

- Soaps and shampoos
- Deodorants
- Air fresheners
- Toothpaste
- Inks and adhesives

Key Messages

- ☑ Always consider poisoning as a DD in any emergency in children.
- ☑ Resuscitation is the main pillar in the management of any poisoning in saving life.
- ☑ All antidotes must be available in ED.
- ☑ Always take the help from Poison Information Center.

Contact details of Poison Information Center:

- ☑ All India Institute of Medical Sciences: 011-26850691, 26159391, 265936772.
- ☑ Poison Information Center: Amrita Institute of Medical Sciences. Kochi +91-484-2858056 (direct) +91-484-2851234; Ext-8056, 6034, 8096, 09895282388
- ☑ *CMC Vellore*: 1800-425-1213

Indian Academy of Pediatrics (IAP)

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Unprovoked Seizures

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